

updatebackground

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1 This program updates the background plasma parameters and DEGAS 2 sources in iterations with a fluid plasma code

\$Id: f39b72edf299c2c89c4a9dbda3fc57a1427da865 \$

This program is designed to work in a manner analogous to *readbackground* when used in conjunction with UEDGE data files. In fact, *updatebackground* can be used in lieu of *readbackground* even on an initial UEDGE-based DEGAS 2 run. The additional functionality arises on subsequent iterations. In these cases, the code looks for the *oldsourcefile* (specified in the *degas2.in*) file. This will have been generated by *readbackground* or *updatebackground* prior to the previous DEGAS 2 run and provides a record of the sources used in that run. The magnitude of these sources will be compared with the current ones. If the relative weights of the sources are “similar” to the originals (“similar” is controlled by the parameters *so_rel_wt_min* and *so_rel_wt_max*), the changes are accounted for with weighting factors. If the weight changes are larger, the source sampling probabilities are recomputed.

```
"updatebackground.f" 1 ≡  
  @m FILE 'updatebackground.web'
```

The unnamed module.

```
"updatebackground.f" 1.1 ≡  
  ⟨ Functions and Subroutines 1.2 ⟩
```

The main program.

```
⟨ Functions and Subroutines 1.2 ⟩ ≡  
  program updatebackground  
    implicit_none_f77  
    implicit_none_f90  
    character*FILELEN datafilename  
    call command_arg(1, datafilename)  
    call ud_bk_main(datafilename)  
    stop  
  end
```

See also sections 1.3 and 1.4.

This code is used in section 1.1.

§1.3 [#4] **updatebackground** This program updates the background plasma parameters and DEGAS 2 sources in iteration

The main subroutine.

⟨Functions and Subroutines 1.2⟩ +=

```

subroutine ud_bk_main(datafilename)
  implicit none f77
  zn_common
  pr_common
  bk_common
  implicit none f90

  character*FILELEN datafilename    // Input

  integer update, uedge, geom_modified    // Local

  ⟨Memory allocation interface 0⟩    /* As in readbackground, generalize the UEDGE interface to
    permit additional options or information to be specified in a text file. To use just a UEDGE
    data file by itself, the user would have to give it a “.u” extension. */

  @#if 0
    uedge = FALSE
    if (index(datafilename, ' .u') > 0) then
      uedge = TRUE
    else
      assert('File_name_must_end_in_.u' ≡ ' ')
    end if
  @#endif

  uedge = TRUE
  call readfilenames

  call read_geometry

  call nc_read_elements
  call nc_read_species
  call nc_read_materials
  call nc_read_reactions
  call nc_read_pmi
  call nc_read_problem

  call setup_back_arrays(geom_modified)
  assert(geom_modified ≡ FALSE)    /* Try reading old sources file. If present, assume that this run
    is an update ⇒ update = TRUE. */
  call nc_read_old_sources(update)

  if (update ≡ TRUE) then
    call update_uedge_background(datafilename, update)
  else
    call read_uedge_background(datafilename, update)
    call set_background_sources
    call init_wt_alias(uedge)
  end if

  call write_background

  return
end

```

§1.4 [#5] **updatebackground** This program updates the background plasma parameters and DEGAS 2 sources in iteration

Calls routines to update the background plasma data. Also reads in old source information to use in deciding how to update the sources for the next run.

⟨ Functions and Subroutines 1.2 ⟩ +=

```
subroutine update_uedge_background(datafilename, update)

    define_varp(old_current, FLOAT, source_seg_ind)
    define_varp(old_rel_wt, FLOAT, source_seg_ind)
    define_varp(old_tot_curr, FLOAT, source_grp_ind)

    implicit_none_f77
    pr_common      // Common
    sp_common
    so_common
    bk_common
    implicit_none_f90

    integer update      // Input
    character*FILELEN datafilename

    integer old_grps, old_seg_tot, grp, i_seg, jr

    declare_varp(old_current)
    declare_varp(old_rel_wt)
    declare_varp(old_tot_curr)

    st_decls

    ⟨ Memory allocation interface 0 ⟩

    var_alloc(old_current)
    var_alloc(old_rel_wt)
    var_alloc(old_tot_curr)    /* Note: could update weights based only on last set of currents (using
                                the old weights as well), but then we couldn't be sure of the total change since the original
                                probability aliases were set. Hence, we really do want to dump out the source data for rereading
                                here only when the aliases are updated. */

    old_grps = so_grps
    old_seg_tot = so_seg_tot
    do grp = 1, so_grps
        old_tot_curr_grp = so_tot_curr(grp)
    end do
    do i_seg = 1, so_seg_tot
        old_current_i_seg = source_current_i_seg
        old_rel_wt_i_seg = source_segment_rel_wt_i_seg
    end do
    assert(update == TRUE)
    call read_uedge_background(datafilename, update)

    if (pr_bkrc_num > 0) then
        do jr = 1, pr_bkrc_num
            so_grps++
            so_seg_tot = so_base(so_grps) - 1
            call set_background_rate(jr, so_grps, update)
        end do
    end if

    assert(so_grps == old_grps)
    assert(so_seg_tot == old_seg_tot)
```

§1.4 [#5] **updatebackground** This program updates the background plasma parameters and DEGAS 2 sources in iteration

```
    call update_wt_alias(old_current, old_rel_wt, old_tot_curr)  
  
    var_free(old_current)  
    var_free(old_tot_curr)  
    var_free(old_rel_wt)  
  
    return  
end
```

assert: 1.3, 1.4.

bk_common: 1.3, 1.4.

command_arg: 1.2.

datafilename: 1.2, 1.3, 1.4.

declare_varp: 1.4.

define_varp: 1.4.

degas2: 1.

FALSE: 1.3.

FILE: 1.

FILELEN: 1.2, 1.3, 1.4.

FLOAT: 1.4.

geom_modified: 1.3.

grp: 1.4.

i_seg: 1.4.

implicit_none_f77: 1.2, 1.3, 1.4.

implicit_none_f90: 1.2, 1.3, 1.4.

in: 1.

index: 1.3.

init_wt_alias: 1.3.

jr: 1.4.

nc_read_elements: 1.3.

nc_read_materials: 1.3.

nc_read_old_sources: 1.3.

nc_read_pmi: 1.3.

nc_read_problem: 1.3.

nc_read_reactions: 1.3.

nc_read_species: 1.3.

old_current: 1.4.

old_grps: 1.4.

old_rel_wt: 1.4.

old_seg_tot: 1.4.

old_tot_curr: 1.4.

oldsourcefile: 1.

pr_bkrc_num: 1.4.

pr_common: 1.3, 1.4.

read_geometry: 1.3.

read_uedge_background: 1.3, 1.4.

readbackground: 1, 1.3.

readfilenames: 1.3.

set_background_rate: 1.4.
set_background_sources: 1.3.
setup_back_arrays: 1.3.
so_base: 1.4.
so_common: 1.4.
so_grps: 1.4.
so_rel_wt_max: 1.
so_rel_wt_min: 1.
so_seg_tot: 1.4.
so_tot_curr: 1.4.
source_current: 1.4.
source_grp_ind: 1.4.
source_seg_ind: 1.4.
source_segment_rel_wt: 1.4.
sp_common: 1.4.
st_decls: 1.4.

TRUE: 1.3, 1.4.

ud_bk_main: 1.2, 1.3.
uedge: 1.3.
update: 1.3, 1.4.
update_uedge_background: 1.3, 1.4.
update_wt_alias: 1.4.
updatebackground: 1, 1.2.

var_alloc: 1.4.
var_free: 1.4.

write_background: 1.3.

zn_common: 1.3.

⟨ Functions and Subroutines 1.2, 1.3, 1.4 ⟩ Used in section 1.1.

⟨ Memory allocation interface 0 ⟩ Used in sections 1.4 and 1.3.

COMMAND LINE: "fweave -f -i! -W[-ybs15000 -ykw800 -ytw40000 -j -n/
/Users/dstotler/degas2/src/updatebackground.web".

WEB FILE: "/Users/dstotler/degas2/src/updatebackground.web".

CHANGE FILE: (none).

GLOBAL LANGUAGE: FORTRAN.